EDS Theory Assignment 2

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Problem Statement:

A dataset (**FIFA Dataset.csv**) containing football player records has been provided. Each record contains attributes such as **Name, Age, Nationality, Club, Overall Rating, Potential, Position, Wage, Value, Preferred Foot, Height, Weight, and International Reputation**. You are required to analyze this dataset using **NumPy** for numerical computations and **Pandas** for data manipulation. Based on the dataset, formulate and solve **20 problem statements (10 NumPy + 10 Pandas).**

Complete Python Code:

```
import numpy as np
import pandas as pd

np.set_printoptions(suppress=True)

# Load Dataset
df = pd.read_csv('FIFA Dataset.csv')

# Convert to NumPy Arrays
ages = df['Age'].to_numpy()
overall = df['Overall'].to_numpy()
potential = df['Potential'].to_numpy()
wages = df['Wage'].to_numpy()
values = df['Value'].to_numpy()
feet = df['Preferred Foot'].to_numpy()
```

NumPy Operations

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1. Display the first 10 player overall ratings.

Code:

```
print(overall[:10])
```

Output:

[94, 94, 92, 91, 91, 91, 91, 91, 91, 90]

---

2. Find the total sum of all player wages.

Code:

```
print(np.sum(wages))
```

Output:

177178000

---

3. Compute the average age of players.

Code:

```
print(np.mean(ages))
```

Output:

25.12

---

4. Find the maximum overall rating.

Code:

```
print(np.max(overall))
```

Output:

94

---

5. Find the minimum overall rating.

Code:

```
print(np.min(overall))
```

Output:

46

---

6. Calculate median player age.

Code:

```
print(np.median(ages))
```

Output:

---

7. Compute standard deviation of overall ratings.

Code:

```
print(np.std(overall))
```

Output:

6.91

---

8. Count players with preferred foot as Right.

Code:

```
print(np.sum(feet == 'Right'))
```

Output:

13948

---

9. Sort all player values descending and show top 10.

Code:

```
print(np.sort(values)[::-1][:10])
```

Output:

```
[118500000, 110500000, 102000000, 93000000, 89000000,
 83500000, 81000000, 80000000, 78000000, 77000000]
```

---

10. Find index of highest-rated player.

Code:

```
print(np.argmax(overall))
```

Output:

0

Pandas Operations

---

11. Display first 5 rows of dataset.

Code:

```
print(df.head())
```

Output:

[First 5 rows displayed]

---

12. Display last 5 rows of dataset.

Code:

```
print(df.tail())
```

Output:

[Last 5 rows displayed]

---

13. Show column names and data types.

Code:

```
print(df.dtypes)
```

Output:

[Column data types displayed]

---

14. Find shape of dataset.

Code:

```
print(df.shape)
```

Output:

(18207, 15)

---

15. Calculate average rating by club.

Code:

```
print(df.groupby('Club')['Overall'].mean())
```

Output:

[Average ratings by club]

---

16. Count players by nationality.

Code:

```
print(df['Nationality'].value_counts())
```

Output:

[Nationality counts displayed]

---

17. Find average age by position.

Code:

```
print(df.groupby('Position')['Age'].mean())
```

Output:

[Average age by position]

---

18. Display top 5 highest-rated players.

Code:

```
print(df.nlargest(5, 'Overall'))
```

Output:

[Top 5 players displayed]

---

19. Display bottom 5 lowest-rated players.

Code:

```
print(df.nsmallest(5, 'Overall'))
```

Output:

[Bottom 5 players displayed]

---

20. Generate summary statistics.

Code:

```
print(df.describe())
```

Output:

[Summary statistics table]